Generating Functions. (chap 4.

Recall the recursively defined sequence of integers, the fibonacci numbers.

$$f_0 = 0 \quad f_1 = 1 \quad \forall i \geq 2 \quad f_i = f_{i-2} + f_{i-1}.$$

We want a closed form for these

Once nice way is using Generating functions.

<u>Def</u> The <u>generating function</u> $g(x)$ of a sequence

$$h_0, h_1, h_2, \ldots h_n, \ldots$$

is the power series

$$g(x) = h_0 + h_1 x + h_2 x^2 + \ldots$$

We treat this as a formal construction, so don't worry about such things as convergence.

Observe that if we have $g(x)$ then $g^{(i)}(0) = h_i / i!$, so $g(x)$ is uniquely determined by $h_0 .. h_n ..$, and contains all information about it.

The generating function allows compact representation of a sequence

<u>Ex</u> The sequence 1, 1, 1, 1 ...
has generating function

$$g(x) = 1 + x + x^2 + x^3 + \dots = \frac{1}{1-x}.$$

Aside.

Where does $1/1-x$ com from, recall that where

$$1 + x^c + x^{2c} + x^{3c} + x^{4c} = S.$$

$$S - x^c S = 1 - x^{5c} \text{ so } S = \frac{1-x^{5c}}{1-x^c}$$

and where

$$1 + x^c + x^{2c} + \dots = S.$$

$$S - x^c S = 1 \quad \text{so} \quad S = \frac{1}{1-x^c}.$$

Now lets look at the coefficients in

$$\left(\frac{1}{1-x}\right)^3 = (1 + x + x^2 + \dots)^3.$$

A x^n comes from an element x^a in the first factor x^b in the second and x^c in the third, satisfying

$$a + b + c = n$$

$a, b, c \geq 0.$

So the generating function nicely describeds our combination with replacement problem. And the nth cofficient is $\binom{n+2}{2}$. So n^{th} coeff of $\left(\frac{1}{1-x}\right)^k = \binom{n+k-1}{k-1}$.

We can deal with restrictions to

<u>Ex</u> How many solutions are there to

$$e_1 + e_2 + e_3 = n.$$

where

$$0 \le e_1 \le 4, \quad e_2 = 1 + 5c \quad e_3 = 0 \text{ or } 1$$

This is the nth coefficient of

$$g(x) = (1 + x + \cdots + x^4)(x + x^6 + x^{11} + \cdots)(1 + x)$$

$$= \frac{1-x^5}{1-x} \cdot x \frac{1}{1-x^5} \cdot 1 + x$$

$$= \frac{1}{1-x}(1+x)x$$

$$= (1 + x + x^2 + \cdots)(x^2 + x)$$

So $g(0) = 0$
$g(1) = 1$
$g(n) = 2 \quad n \ge 2.$

<u>Ex</u> $(1+x)^n$ generates the sequence

$$h_0 = \binom{n}{0} \quad h_1 = \binom{n}{1} \quad h_i = \binom{n}{i}.$$

Now we use generating functions to get a closed form for the fibonacci sequence

$$f_0 = 0 \quad f_1 = 1 \quad f_i = f_{i-2} + f_{i-1}.$$

First we find a generating function:

Let $g(x) = f_0 + f_1 x + f_2 x^2 + \dots$

From $f_i = f_{i-1} + f_{i-2}$. we get

$$\begin{array}{lcl} g(x) & & f_0 + x f_1 + x^2 f_2 + \dots \\ -x\,g(x) & = & -(\quad x f_0 + x^2 f_1 + \quad) \\ -x^2 g(x) & & -(\quad x^2 f_0 + \quad) \\ & = & f_0 + x(f_1 - f_0) + 0. \end{array}$$

So $g(x)(1 - x - x^2) = f_0 + x(f_1 - f_0)$
$= x.$

Thus $g(x) = \dfrac{x}{1 - x - x^2}$.

Now $0 = (1 - x - x^2) = (1 - x/2)^2 - \frac{5x^2}{4}$
$\Rightarrow (1 - x/2) = \pm \frac{x\sqrt{5}}{2}$
$\Rightarrow 1 = x\left(\frac{1 \pm \sqrt{5}}{2}\right)$

So $(1 - x - x^2) = \left(1 - x\left(\frac{1-\sqrt{5}}{2}\right)\right)\left(1 - x\left(\frac{1+\sqrt{5}}{2}\right)\right)$.

Using partial fractions to decompose

$$g(x) = \frac{x}{(1-x\frac{1+\sqrt{5}}{2})(1-x\frac{1-\sqrt{5}}{2})}.$$

We set it equal to

$$\frac{c_1}{1-x\frac{1+\sqrt{5}}{2}} + \frac{c_2}{1-x\frac{1-\sqrt{5}}{2}}$$

and get

$$x = c_1\left(1-x\frac{1-\sqrt{5}}{2}\right) + c_2\left(1-x\frac{1+\sqrt{5}}{2}\right).$$

so $c_1 + c_2 = 0.$ $\Rightarrow$ $c_1 = -c_2$

and
$$\begin{aligned} 1 &= -c_1\frac{1-\sqrt{5}}{2} - c_2\frac{1+\sqrt{5}}{2} \\ &= c_2\frac{1-\sqrt{5}}{2} - c_2\frac{1+\sqrt{5}}{2} \\ &= -\sqrt{5}\,c_2. \end{aligned}$$

So $c_2 = 1/-\sqrt{5}$ $c_1 = \frac{1}{\sqrt{5}}$, and

$$\begin{aligned} g(x) &= \frac{1}{\sqrt{5}}\left(\frac{1}{1-x\frac{1+\sqrt{5}}{2}}\right) - 1/\sqrt{5}\left(\frac{1}{1-x\frac{1-\sqrt{5}}{2}}\right). \\ &= 1/\sqrt{5}\left(1 + \left(\frac{1+\sqrt{5}}{2}\right)x + \left(\frac{1+\sqrt{5}}{2}\right)^2 x^2 + \right) \\ &\quad - 1/\sqrt{5}\left(1 + \left(\frac{1-\sqrt{5}}{2}\right)x + \left(\frac{1-\sqrt{5}}{2}\right)^2 x^2 + \dots\right) \end{aligned}$$

Reading off the nth coefficient

$$f_n = 1/\sqrt{5}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right).$$